



Lab: Using Generative AI with Excel, SQL, Power BI, and Tableau

Estimated time: 30 minutes

Learning objectives:

After completing this lab, you will be able to:

- Generate clear data summaries for non-technical stakeholders using AI
- Create SQL queries from business questions using AI assistance
- Design dashboard specifications for tools such as Excel, Power BI, and Tableau

Introduction

In this lab, you will explore how to use AI for common data analysis tasks. You will create executive summaries from dataset descriptions, generate SQL queries to answer business questions, and design dashboard specifications for visualization tools. These skills help you quickly transform raw data into actionable insights that stakeholders can understand and use for decision-making.

Required tools

- Web AI tool (ChatGPT or Claude)
- Sample dataset description or metadata

Exercise 1: Generate an executive data summary

In this exercise, you will use AI to create a clear, concise summary of a dataset for stakeholders who may not have technical expertise. You will translate technical data descriptions into plain language that explains what the data contains, what insights it provides, and how it can support business decisions.

Step 1: Analyze dataset description

Obtain or create a sample dataset description. For example, sales transactions, customer data, inventory records, and website analytics.

Identify key information such as data sources, column names and types, time period covered, and any known data quality issues.

Step 2: Generate stakeholder summary using AI

Open your AI tool and provide the dataset description.

Prompt the AI to create a non-technical summary explaining what the data shows and how it can be used.

Review the summary to ensure that it avoids jargon and clearly communicates business value.

Hint

Example prompt: "Create an executive summary of this dataset for business stakeholders. Explain: what data is included (in plain language), what time period it covers, what business questions it can answer, what key insights are available, and how this data supports decision-making. Use simple language without technical jargon. Keep the summary to 3–4 paragraphs."

Exercise 2: Generate SQL queries from business questions

In this exercise, you will use AI to convert business questions into SQL queries. Rather than writing SQL from scratch, you will describe what you want to know in plain English and have AI generate the appropriate SQL code. This approach speeds up data analysis and helps you get answers faster.

Step 1: Define business questions

Based on your dataset, write 3–4 business questions you need to answer. For example, "What is our total revenue by month?", "Which products have the highest sales?", and "What is the average order value by customer segment?".

Ensure that questions are specific and reference metrics that exist in your dataset.

Step 2: Generate SQL queries using AI

For each business question, prompt the AI to write an SQL query.

Provide the AI with table and column names from your dataset to ensure accuracy.

Review each generated query to verify it matches your dataset structure.

Hint

Example prompt: "Write an SQL query to answer this question: [your question]. Use this table structure: [paste table schema with column names and types]. Include appropriate GROUP BY, ORDER BY, and aggregate functions (SUM, AVG, COUNT) as needed. Add comments explaining what each part of the query does."

Sample query structure: SELECT [columns], [aggregate functions] FROM [table] WHERE [filters] GROUP BY [grouping columns] ORDER BY [sort columns] LIMIT [number].

Exercise 3: Design dashboard specification for Power BI, Tableau, or Excel

In this exercise, you will use AI to plan a dashboard that visualizes your data insights. You will determine which charts and visualizations best answer stakeholder questions, then create a specification that could be implemented in Power BI, Tableau, or Excel. This planning step ensures that your dashboard will effectively communicate insights before you invest time building it.

Step 1: Identify dashboard objectives

List the 3–5 most important questions your dashboard should answer.

Identify your target audience and their level of technical expertise.

Determine if the dashboard is for a high-level overview or a detailed analysis.

Step 2: Generate visualization recommendations using AI

Prompt the AI to recommend appropriate chart types for each question.

Ask the AI to suggest a dashboard layout and organization.

Request recommendations for filters and interactive elements.

Step 3: Document dashboard specification

Create a specification document that includes the dashboard title and purpose, a list of visualizations with chart types, data sources for each visual, and interactive filters or controls.

Organize visualizations by priority, with the most important ones at the top.

Hint

Example prompt: "Design a dashboard specification to answer these questions: [list questions]. For each question, recommend: the best visualization type (bar chart, line chart, pie chart, table, KPI card, etc.), what data to display on each axis or dimension, whether to use filters or drill-downs, and how to arrange the dashboard layout. Consider that this will be built in [Power BI/Tableau/Excel]. Target audience: [describe audience]."

Visualization guidelines: Bar charts for comparisons | Line charts for trends over time | Pie charts for proportions (use sparingly) | Tables for detailed data | KPI cards for single metrics | Heat maps for patterns across two dimensions.

Summary

In this lab, you used AI to create data summaries for stakeholders, generate SQL queries from business questions, and design dashboard specifications for visualization tools. These skills help you work more efficiently with data analysis tools and communicate insights effectively to different audiences.